

HALLHAXASEN, Sweden

WMO: 022360

Lat: 63.7685N

Lon: 15.3279E

Elev: 376

StdP: 96.89

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-20.5	-25.9	0.4	-23.2	-22.9	0.5	-20.1	8.2	0.6	6.8	-0.3	1.0	40	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	24.6	16.1	22.7	15.4	20.6	14.4	17.5	22.3	16.4	20.6	15.3	19.3	2.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.6	19.1	14.6	10.8	18.0	13.6	10.1	17.1	50.6	22.4	47.2	20.4	44.2	19.2	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.8	5.1	DB	-26.7	26.9	3.3	2.0	-29.1	28.3	-31.1	29.5	-32.9	30.6	-35.4	32.0	(4)
(5)			WB	-26.9	19.1	3.2	1.4	-29.2	20.1	-31.0	20.9	-32.8	21.6	-35.1	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-8.0	-7.1	-3.5	2.0	7.4	11.5	14.4	12.9	8.1	2.1	-3.0	-6.2	(6)	
	DBStd	9.00	6.24	6.16	4.69	3.45	4.06	3.70	3.34	3.18	3.00	3.72	4.49	6.11	(7)	
	HDD10.0	3044	557	480	419	239	105	24	3	8	71	246	390	502	(8)	
	HDD18.3	5753	816	713	677	488	340	208	129	172	306	504	640	760	(9)	
	CDD10.0	343	0	0	0	0	23	69	139	97	14	1	0	0	(10)	
	CDD18.3	12	0	0	0	0	0	2	7	2	0	0	0	0	(11)	
	CDH23.3	111	0	0	0	0	3	26	66	16	0	0	0	0	(12)	
	CDH26.7	8	0	0	0	0	0	0	7	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.2	1.9	2.1	2.6	2.5	2.6	2.5	2.1	1.9	2.1	2.2	1.9	1.9	(14)	
(15) Precipitation	PrecAvg	589	36	27	27	33	47	60	87	65	62	50	47	39	(15)	
	PrecMax	791	69	49	44	59	92	99	189	146	137	185	96	90	(16)	
	PrecMin	403	8	8	8	10	14	11	29	4	15	0	24	10	(17)	
	PrecStd	110	16	12	11	13	23	22	48	42	30	39	22	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	5.3	8.1	16.7	23.6	25.8	27.8	25.3	18.7	13.4	8.7	5.1	(19)	
		MCWB	3.0	3.3	3.9	8.9	13.9	15.4	18.5	17.1	13.9	10.1	7.7	3.8	(20)	
	2%	DB	3.1	4.0	5.7	13.0	20.5	23.7	25.2	23.0	16.4	10.2	5.7	3.6	(21)	
		MCWB	1.9	2.3	3.0	6.8	12.0	14.5	17.0	16.2	12.7	8.3	5.1	2.4	(22)	
	5%	DB	1.4	2.5	4.3	10.3	17.8	20.8	23.1	20.7	14.5	8.8	3.8	2.4	(23)	
		MCWB	0.6	1.3	2.1	5.3	10.9	13.3	16.1	15.3	11.8	7.4	3.2	1.5	(24)	
	10%	DB	0.0	0.7	3.1	8.0	14.8	18.5	20.9	18.7	13.0	7.6	2.5	1.1	(25)	
		MCWB	-0.6	-0.3	1.2	4.2	9.6	12.4	15.3	14.5	10.7	6.4	1.9	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.6	4.7	9.1	14.7	17.2	19.4	18.5	14.9	10.6	7.6	3.9	(27)	
		MCDB	4.4	4.9	7.1	15.1	20.6	23.2	25.4	22.6	17.3	12.2	8.5	4.8	(28)	
	2%	WB	2.0	2.5	3.4	7.5	13.2	15.5	17.9	17.1	13.3	8.8	5.2	2.6	(29)	
		MCDB	2.9	3.8	5.3	12.4	18.6	20.9	23.4	21.0	15.5	9.9	5.7	3.3	(30)	
	5%	WB	0.7	1.3	2.4	5.9	11.6	14.2	16.8	16.0	12.2	7.6	3.4	1.6	(31)	
		MCDB	1.3	2.4	4.1	9.4	16.8	19.6	21.6	19.6	14.1	8.7	3.7	2.3	(32)	
	10%	WB	-0.6	-0.2	1.3	4.5	10.0	13.0	15.8	15.0	11.1	6.5	2.0	0.5	(33)	
		MCDB	-0.1	0.6	2.9	7.4	14.4	17.8	20.0	18.1	12.6	7.5	2.4	1.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.1	7.7	8.0	9.3	9.4	9.1	8.7	7.3	5.7	4.4	5.4	(35)	
		MCDBR	5.3	5.8	7.7	12.3	13.3	12.8	12.1	11.4	9.4	7.3	5.1	5.4	(36)	
	5% WB	MCWBR	4.8	5.0	5.7	7.0	6.9	6.1	5.7	5.7	6.3	5.6	4.8	5.0	(37)	
		MCDWR	5.3	5.7	7.3	10.9	12.2	11.3	10.8	9.7	8.7	6.8	5.1	5.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.199	0.264	0.292	0.326	0.331	0.328	0.341	0.330	0.303	0.287	0.354	0.349	(40)	
		taud	2.017	2.230	2.312	2.401	2.483	2.517	2.493	2.513	2.576	2.473	2.817	2.775	(41)	
	Ebn at Noon	Ebn at Noon	412	672	808	857	885	894	869	848	808	659	396	238	(42)	
		Edh at Noon	27	57	80	92	93	92	92	82	63	46	24	16	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.15	0.71	2.05	3.76	4.66	5.28	4.90	3.64	2.12	0.89	0.23	0.07	(44)	
	RadStd	RadStd	0.01	0.10	0.22	0.32	0.43	0.38	0.53	0.32	0.22	0.13	0.04	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-137	N/A	N/A	N/A

Nomenclature: See separate page